

Discipline : BS-Chemistry
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Subject : Organic Chemistry
Topic : Past Paper 2019
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SHORT ANSWERS

Qno 2

(i)

Electronic Transitions:

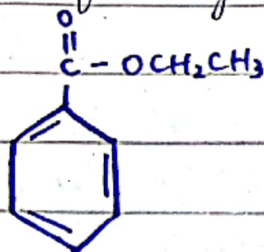
"Electronic transitions refers to the process in which an electron within an atom or molecule changes its energy level by moving from one electronic orbital or energy state to another."

- Electronic transitions occur when an

electron emits or absorb a photon of light.

Electronic Transitions in Ethyl benzoate:

The structure of Ethyl benzoate is



The possible electronic transitions in ethyl benzoate are:

- $\pi - \pi^*$ transitions:

These transitions occur between the π -electrons in the benzene ring and π -electrons in the $C=O$ bond.

- $n - \pi^*$ transitions:

These transitions occur between the n -electrons on the oxygen atom in the $C=O$ bond and the π^* electron in the benzene ring.

- $n - \sigma^*$ transitions:

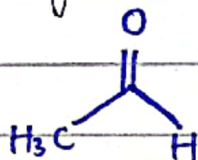
These transitions occur between the π -electrons in the benzene ring and σ^* antibonding orbitals of the $C-O$ and $C-C$ bonds.

• $\sigma - \pi^*$ transitions:

These transitions occur between the σ -bonding orbitals of the C-O and C-C bonds and π^* antibonding orbitals of the benzene ring.

In Acetaldehyde:

The structure of acetaldehyde is



Possible electronic transitions in acetaldehyde are:

• $\pi - \pi^*$ transitions:

These transitions occur between the π -electrons in the C=O bond and π -electrons in the C=C bond.

• $n - \pi^*$ transitions:

These transitions occur between the n -electrons on the oxygen atom in C=O and the π^* electrons in the C=C bond.

• $\pi - \sigma^*$ transitions:

These transitions

occur between the π -electrons in the $C=O$ and $C=C$ σ^* antibonding orbitals of the $C-H$ bonds.

• $\sigma - \pi^*$ transitions:

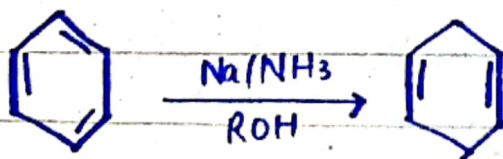
These transitions occur between the (σ ground state of the) σ bonding orbitals of the $C-O$ and $C-C$ bonds and the π^* -antibonding orbitals of the $C=O$ and $C=C$ bonds.

(ii)

Birch Reduction:

The Birch reduction is an organic chemical reaction where aromatic compounds which have a benzenoid ring are converted into 1,4-cyclohexadiene which have two hydrogen atoms attached at opposite ends of the molecule.

Reaction:



Mechanism:

Birch reduction, ^{medium} usually consists of lithium or sodium in liquid ammonia. An alcohol is usually added to serve as a proton source.

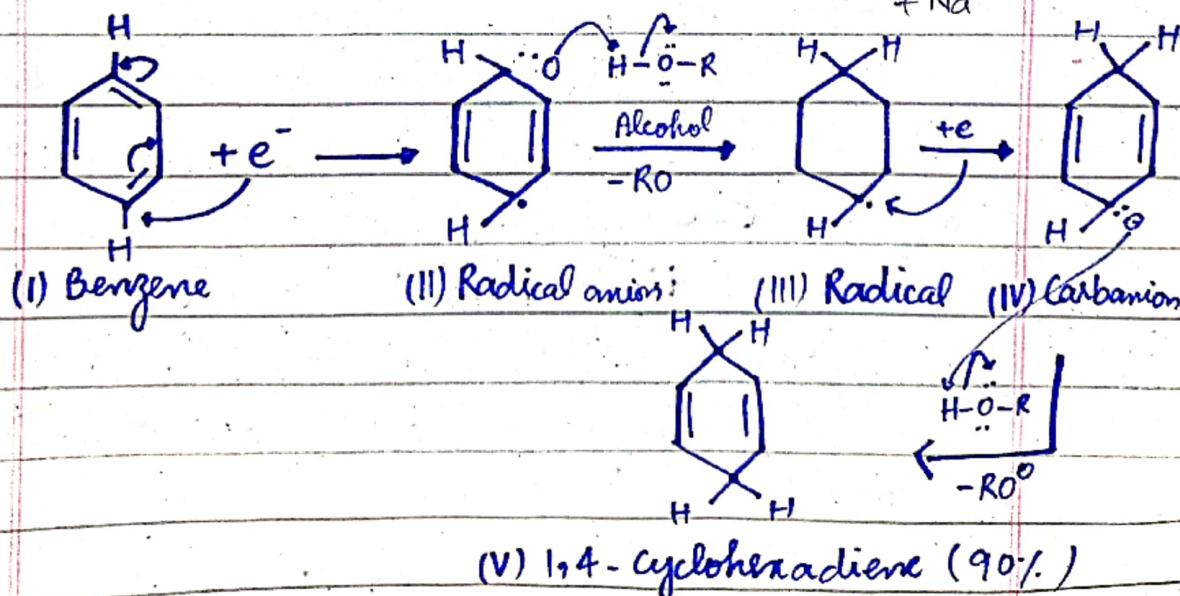
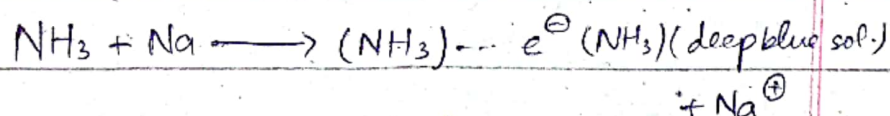
→ The mechanism of this reduction begins the addition of the electron to form a radical anion (II) which is prepared from (I).

- The radical ion is protonated and gives a neutral radical (III).

- Addition of another electron gives a carbanion (IV), which is protonated to yield 1,4-cyclohexadiene (V).

This reaction is called **Birch reduction**.

Mechanism is as follows



The isolated double bonds in the dihydro product are ^{less} much easily reduced than the aromatic ring and thus reduction stops at the dihydro stage.

(iii)

Ethylene does not show $C=C$ stretching band while propylene does:

Ethylene:

Ethylene (C_2H_4) is the simplest alkene. It is a symmetrical molecule, with two identical hydrogen atoms on each carbon atom.

The $C=C$ double bond in ethylene is rigid. This means that the carbon atom cannot rotate freely around the double bond.

When the $C=C$ double bond in the ethylene vibrates, the dipole moment of the molecule does not change. This is because the two carbon atoms are moving in opposite direction, and their dipole moment cancel each other out.

As a result of the lack of change of dipole moment, the $C=C$ stretching vibration in ethylene is **IR-inactive**. This means that ethylene does not show an absorption band in the IR spectrum due to $C=C$ stretch.

Propylene:

Propylene (C_3H_6) is a slightly more complex alkene than ethylene. It has three hydrogen atoms on one of the carbon atoms of the double bond & methyl group attached to one of the carbon atoms of the double bond.

Propylene is not a symmetrical molecule and the $C=C$ double bond in propylene is not as rigid as the $C=C$ in ethylene.

When the $C=C$ in propylene vibrates, the **dipole moment** of the molecule does change. As a result of the change in dipole moment, the $C=C$ stretching vibration in propylene is **IR active**.

This means that propylene shows an absorption band in the IR spectrum due to $C=C$ stretching.

(iv)

Role of Free Radicals in Aging

"Free radicals are highly unstable molecules that have an unpaired electron. They are highly reactive and can damage cells and tissues."

Role in Aging:

Free radicals are produced naturally in the body as a result of ~~unnatural~~ normal metabolism.

As we age, our bodies become less efficient at removing free radicals. This can lead to a build-up of free radicals in the body, which can damage cells and tissues.

→ Damage to DNA:

Free radicals can cause oxidative damage to DNA, leading to **mutations** and **genetic instability**.

This DNA damage can result in the accumulation of mutations, which may

contribute to age related diseases and reduced cellular functions:

→ Protein Damage:

- Free radicals can also damage proteins, impairing their structure and functions.
- Damage to proteins can lead to a variety of problems, including age-related diseases such as Alzheimer's disease and Parkinson's diseases.

→ Inflammation:

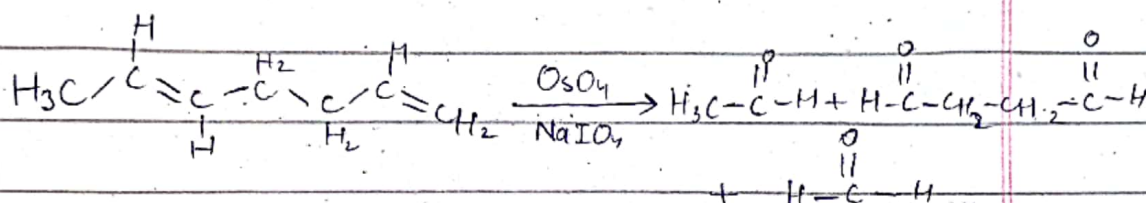
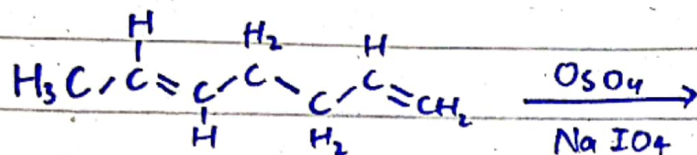
- Free radicals can promote chronic inflammation by activating pro inflammatory pathways.
- Prolonged inflammation is associated with various age-related diseases and contributes to tissue damage.

→ Lipid Peroxidation:

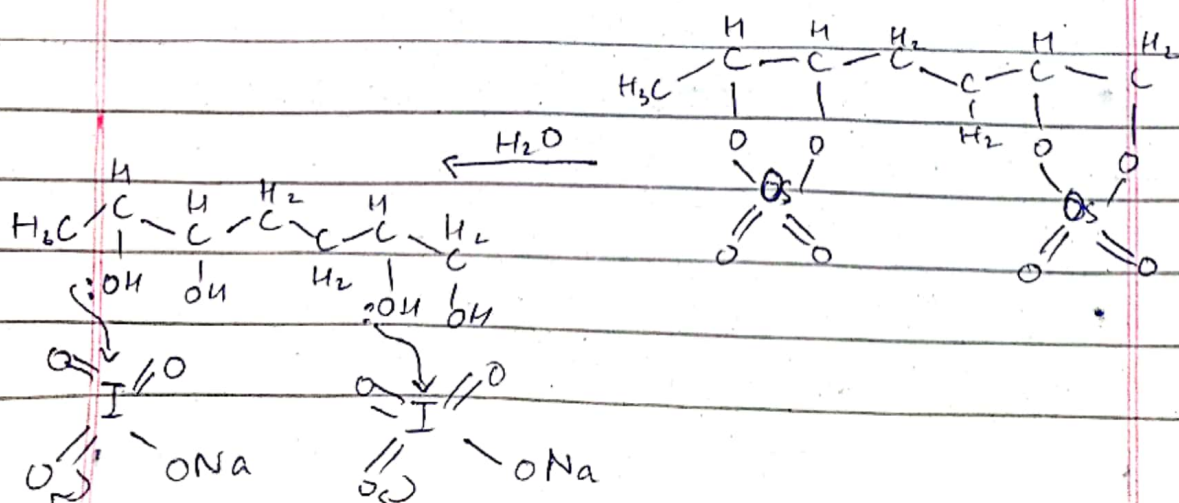
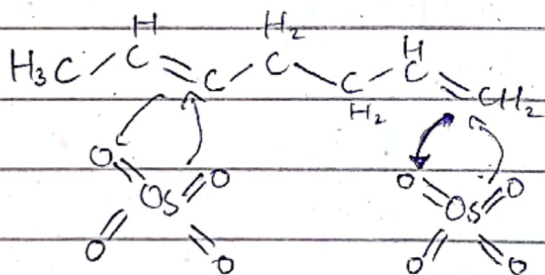
- Lipids in cell membrane are susceptible to oxidation by free radicals through a process called lipid peroxidation.
 - This can compromise the integrity of cell membranes, affecting cellular communication & function.
-

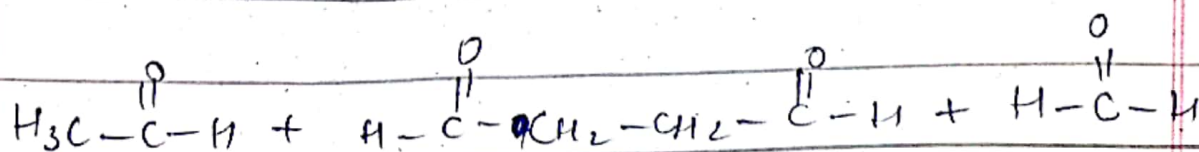
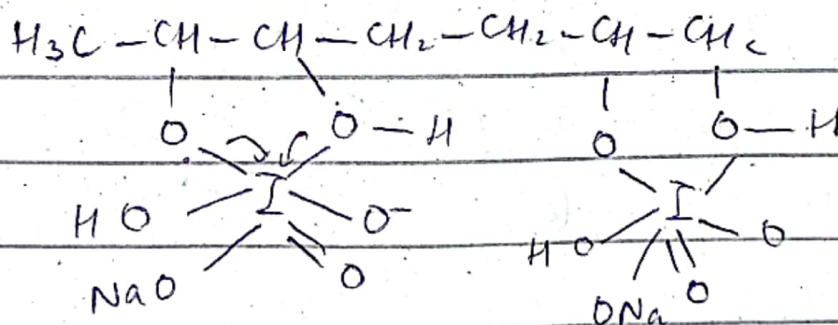
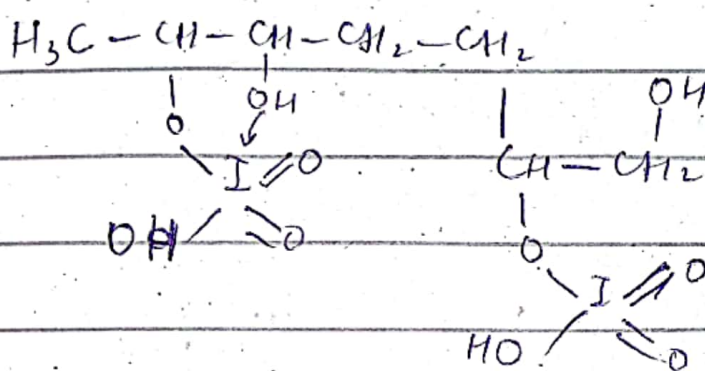
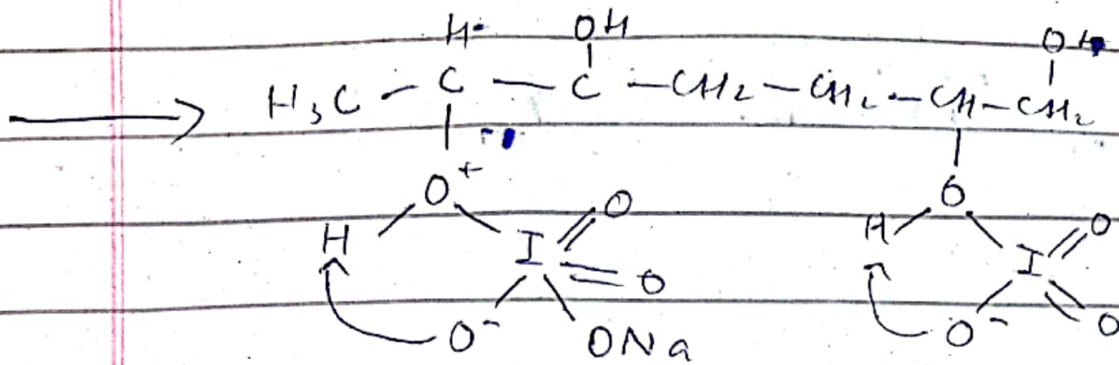
(v)

Write product and mechanism



Mechanism:





LONG ANSWERS

Qno 3

(i)

Swern Oxidation:

Swern oxidation

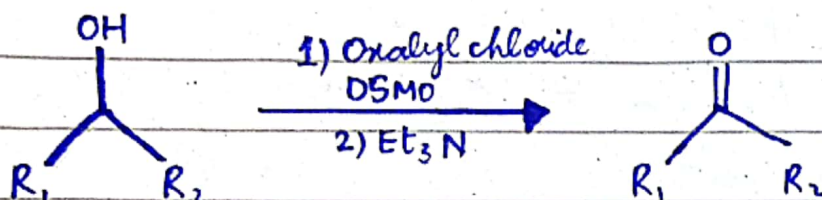
refers to the organic chemical reaction where oxalyl chloride, an organic bases, and dimethyl sulfoxide are employed for the oxidation of primary and secondary alcohols to aldehydes or ketones, respectively.

- It is important to note that these aldehydes do not undergo further reaction to produce carboxylic acids.

- A **disadvantage** of this reaction is the formation of dimethyl sulphide as a side product, which has an unpleasant smell which can become highly disagreeable even in low concentrations. The reaction is an organic redox reaction.

Example,

An example of the Swern reaction is given below:

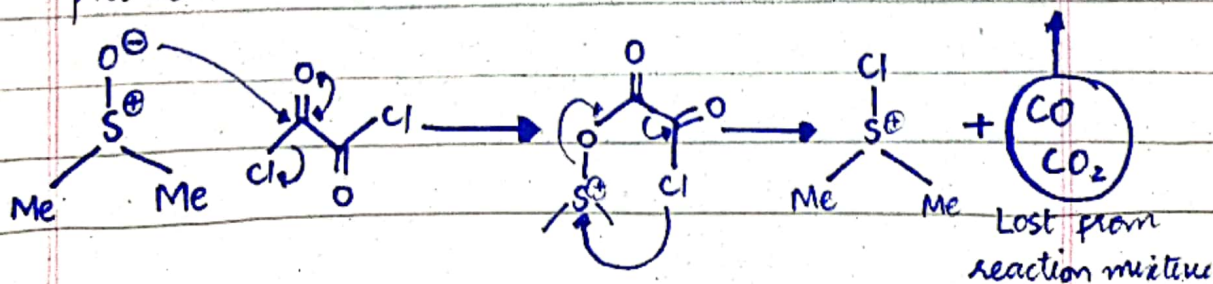


DMSO is the abbreviation used for dimethyl sulfoxide.

Mechanism:

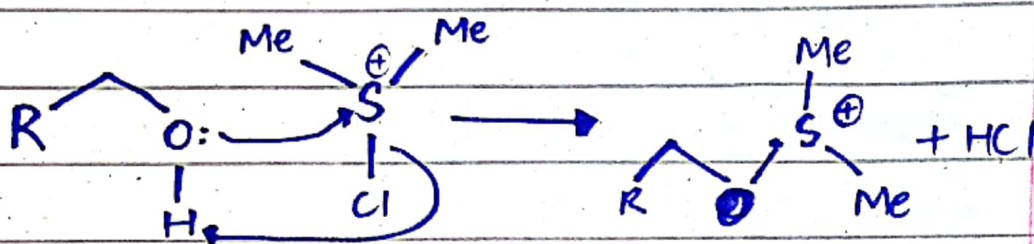
Step 1

A resonance supported dimethyl sulfoxide molecule reacts with oxalyl chloride at low temperatures (below 60°). An intermediate is formed and quickly decomposed to give chloro (dimethyl) sulfonium chloride. Carbon monoxide and dioxide are also formed as by-products.



Step 2

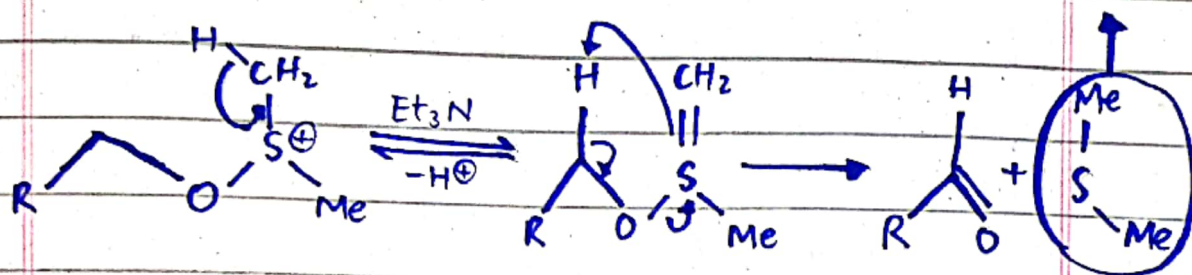
Now an alcohol is added to the chloro (dimethyl) sulfonium chloride. The reaction that ensues gives the required intermediate - the alkoxysulfonium ion along with HCl. This step can be illustrated as -



Step 3

Now a minimum of 2 equivalents of the triethylamine base perform deprotonation of the intermediate alkoxysulfonium ion. This deprotonation gives the neutral dipolar sulfur ylide. This sulfur ylide now decomposes via a ring transition state which is a five-membered. The products formed from this decomposition include the required carbonyl compound and dimethyl sulfide by-product.

This step of the Swern oxidation mechanism is illustrated below.



Thus the required **carbonyl compound** is achieved from the oxidation of the primary (or secondary) alcohol. Tetrahydrofuran or THF can also be used as a solvent for this reaction.

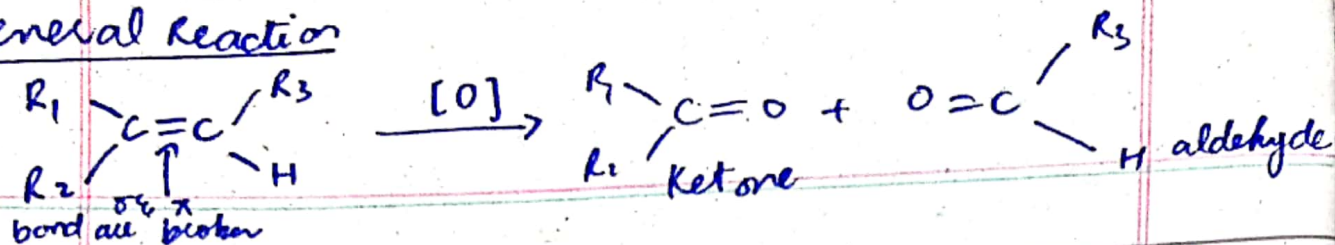
(ii)

Methods for Oxidative Cleavage of double bond:

“Oxidative cleavage of double bonds involve the breaking of a carbon-carbon double bond (C=C) using oxidizing agents.”

→ This reaction convert **alkenes** into **carbonyl compounds**, such as aldehydes & ketones.

General Reaction



Methods

Two widely used methods for cleavage of double bonds are discussed below

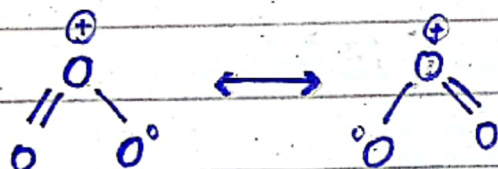
1. Ozonolysis:

"Cleavage with ozone is called ozonolysis."

This is one of the methods of oxidative cleavage. It relies on a two-step procedure using **Ozone (O₃)** as the oxidant in the first step.

Structure of Ozone:

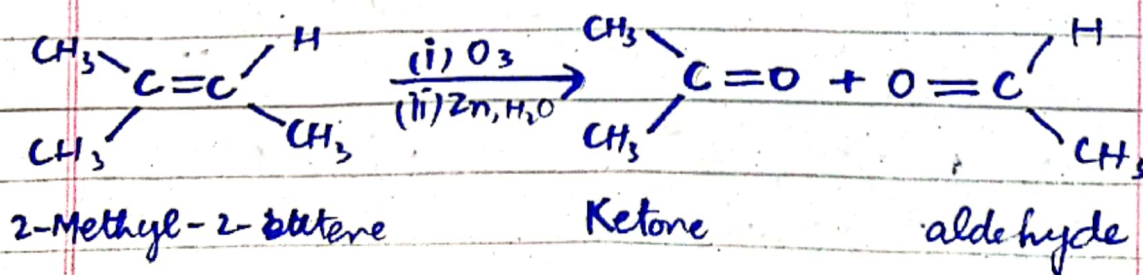
The structure of ozone is

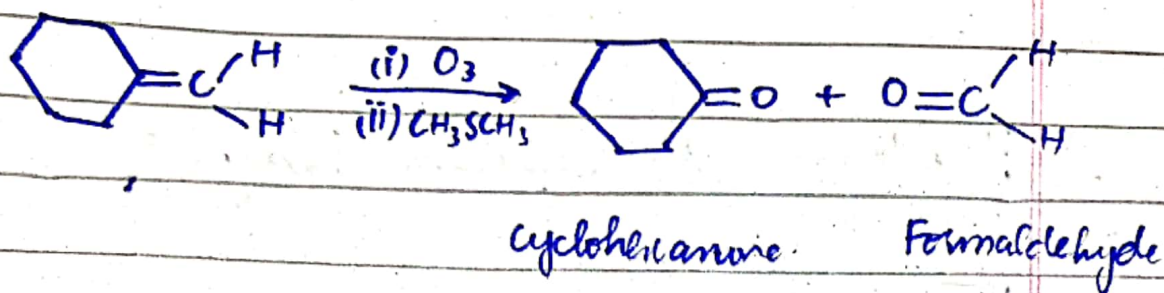


It is highly reactive having 1,3-dipole.

Chemical Reaction:

The overall reaction is as follow -





An unstable intermediate called a **molozone** is produced on addition of O_3 to the π -bond. Molozone then rearranges to an **ozonide**.

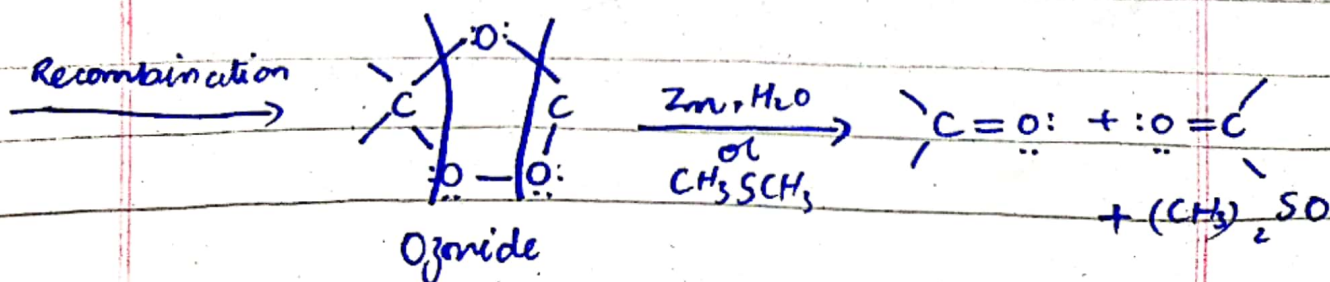
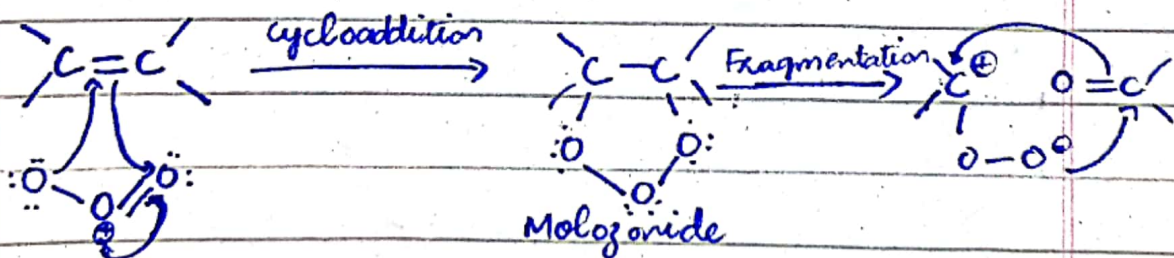
The unstable ozonide is then reduced without isolation to give carbonyl compounds.

The two common reagents used for this purpose are

- Zn in H_2O or dimethyl sulfide (CH_3SCH_3)

Mechanism:

The overall mechanism of reaction of O_3 with alkenes, is given below:



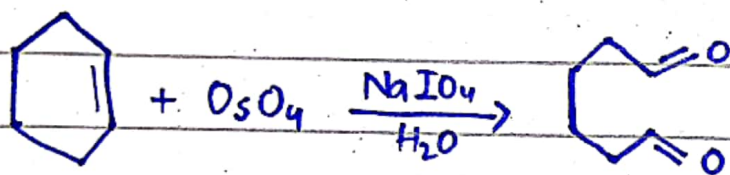
$Zn(OH)_2$ & $(CH_3)_2S=O$ are the by products

Reaction with Lemieux Reagent

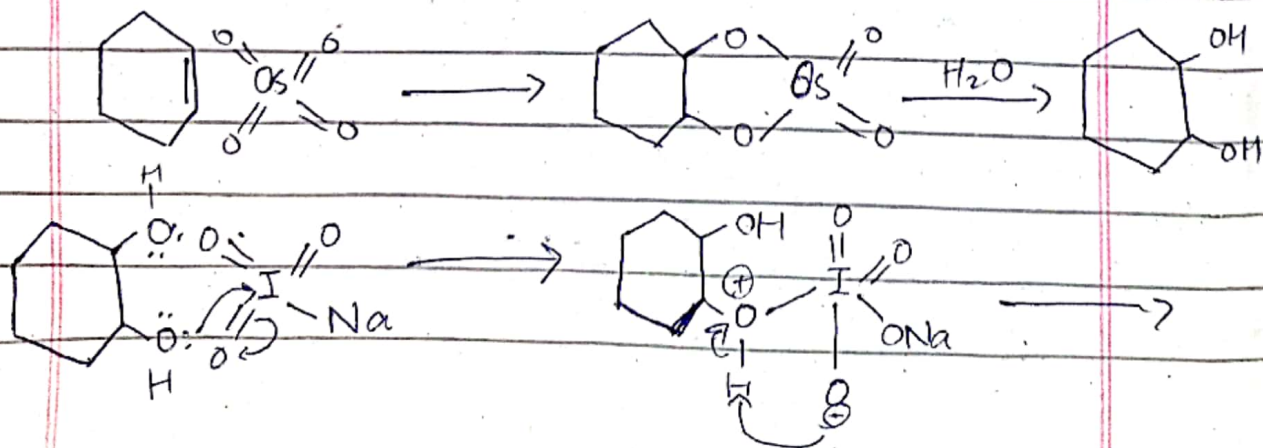
"The conversion of an olefin into two individual aldehydes by means of an oxidative cleavage of a carbon-carbon double bond with osmium tetroxide - sodium periodate is known as the Lemieux-Johnson oxidative cleavage."

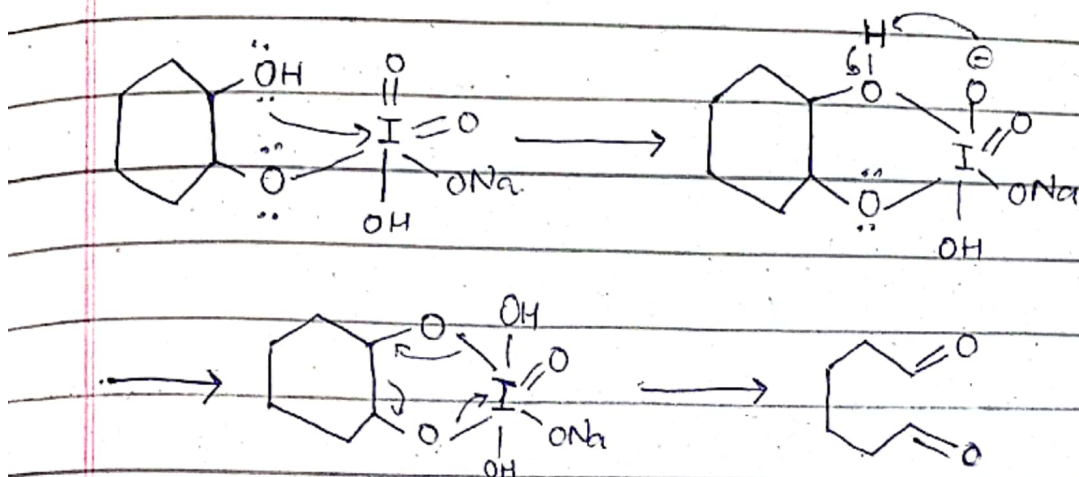
- The combination of osmium tetroxide and sodium periodate is referred to as Lemieux-Johnson reagent.

Reaction



Mechanism





Qno 4

Sample Handling in IR

Infrared (IR) spectroscopy is a powerful technique used to identify and characterize the functional groups present in a molecule. It measures the absorption of infrared radiation by a sample, which causes molecular vibration. The resulting spectrum provides information about the chemical structure of sample.

Sampling methods:

The choice of sampling method for IR spectroscopy depends on the nature of the sample (solid, liquid or gas) and the desired accuracy of

The measurement. Some common sampling methods include:

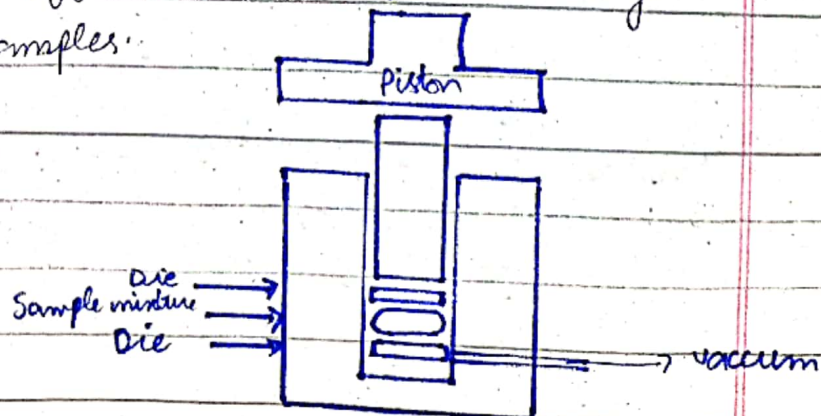
A. Solid Samples:

Solid samples require specific preparation techniques to ensure uniform dispersion and interactions with the infrared beam. Common method include:

1. Pellet Technique:

- The Pellet technique involves mixing a small amount of finely ground sample with an infra-red-transparent salt, typically potassium bromide (KBr). This mixture is pressed into a thin pellet using a hydraulic press.

- The pellet is placed in the path of the IR beam, and the spectrum is recorded. This method is widely used due to its simplicity & effectiveness for a ^{10 tons pressure} wide range of solid samples.



2. Solid Run-in Solution

- Dissolving the solid in a non-aqueous solvent is a convenient method for preparing samples that are soluble in suitable solvents.
- The choice of solvent is crucial as it should not interfere with the IR spectrum of the analyte.
- A drop of the solution is placed on an alkali metal disk, such as sodium chloride or potassium bromide, and the solvent is allowed to evaporate, leaving a thin film of the solute for analysis.
- Alternatively, the solution can be placed in a liquid sample cell, similar to those used for liquid samples.

3. Solid Films:

- Solid films are prepared by depositing a thin layer of the solid sample onto an IR transparent window, typically made of potassium bromide or sodium chloride.
- This method is particularly useful for amorphous solids that cannot be easily pressed into pellets.

- The deposition is often done by evaporating a solution of the solid onto the window, leaving ~~to~~ a thin film behind.

4. Mull Technique :

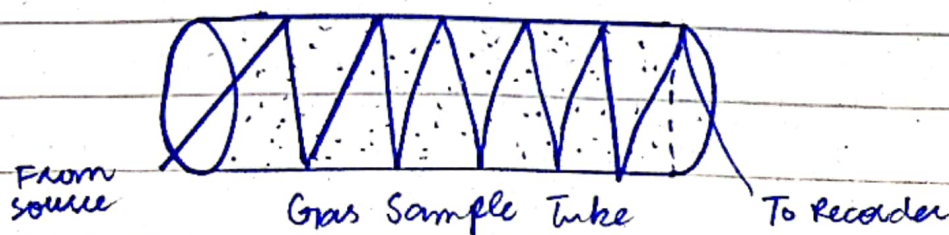
- The mull technique involves mixing a finely ground solid sample with a mulling agent, typically Nujol (mineral oil), to form a thick paste.
- The paste is then spread between two IR-transparent windows, creating a thin film of the sample for analysis.
- Nujol is transparent in most of the IR region, but it does have absorption bands that can interfere with the sample's spectrum.
- This method is primarily used for qualitative analysis, as the mulling agent can affect the quantitative accuracy of the measurement.

B. Sampling of liquids

- Liquid samples are typically analyzed using liquid cells, which are rectangular containers made of IR transparent material such as NaCl, KBr or ThBr.
- The sample is placed between two windows of the cell, and the cell is then placed in the path of IR beam.
- The sample thickness should be adjusted to achieve 15-20% transmittance, ensuring a strong enough signal without saturation.
- For double instruments, matched cells are used to compensate for solvent absorption. One cell contains the sample, while the other contains the pure solvent.
- Cells should be protected from moisture as they are soluble in water. Organic liquid samples should be dried before analysis to avoid interference from water absorption bands.

C. Sampling of solids:

- Gas samples cells are similar to liquid samples cells in terms of construction, with window made of IR transparent material like KBr or NaCl.
- However, gas cells are typically longer (10cm to 1m) to compensate for lower density of gas molecules compared to liquids.
- Multiple reflections can be used to extend the effective path length of the IR beam through the gas sample, increasing sensitivity.
- The gas must not react with the cell windows or reflecting surfaces, as this can affect the measurement.
- IR spectroscopy for gas analysis is less sensitive compared to other methods, and moisture can interfere due to its strong absorption bands.



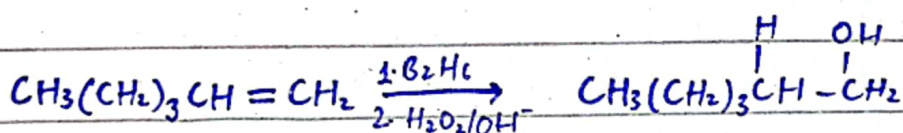
Qno 5

(b)

Hydroboration

“Hydroboration is a two-step reaction that involves the addition of a hydrogen-boron bond (H-B) to an alkene or alkyne, followed by oxidation to produce an alcohol.”

→ It is a versatile and valuable synthetic tool for preparation of alcohols, especially those with anti-Markovnikov regioselectivity



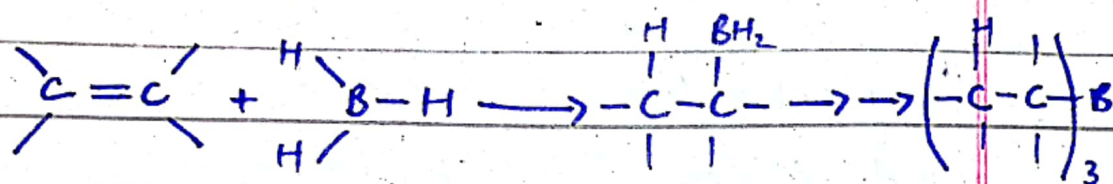
Mechanism:

Its mechanism involves two steps

1. Hydroboration step:

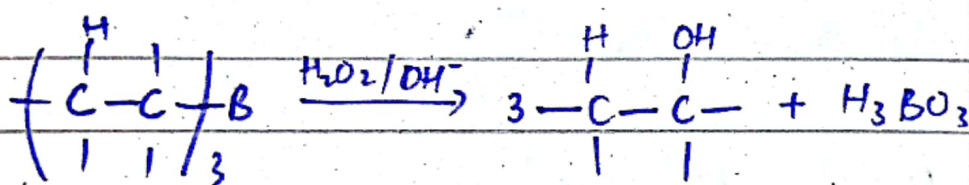
In this step, alkene is treated with diborane, B_2H_6 . Diborane acts as if it were the monomeric species called borane, BH_3 . The compound is usually prepared in an ether solvent such as diethyl

ether or tetrahydrofuran. It adds to the carbon-carbon double bond of one alkene and then adds successively to two or more alkenes to produce a trialkylborane, R_3B . These steps are hydroboration reactions.



2-Oxidation Step:

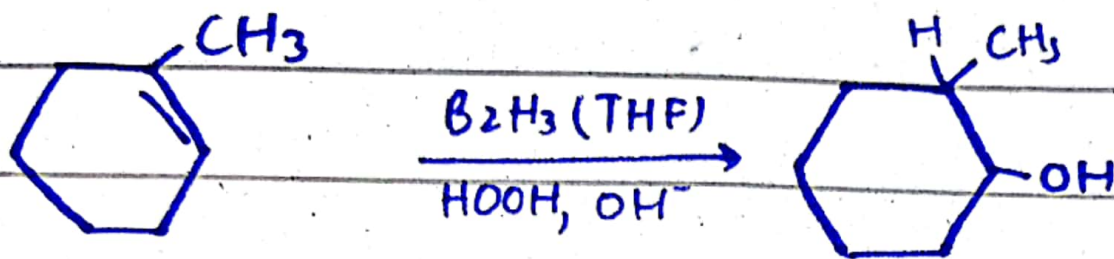
In the oxidation step, the trialkylborane is treated with hydrogen peroxide and base to oxidize the organoborane to form an alcohol.



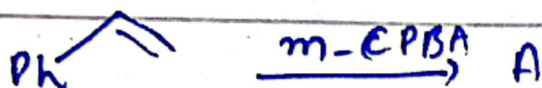
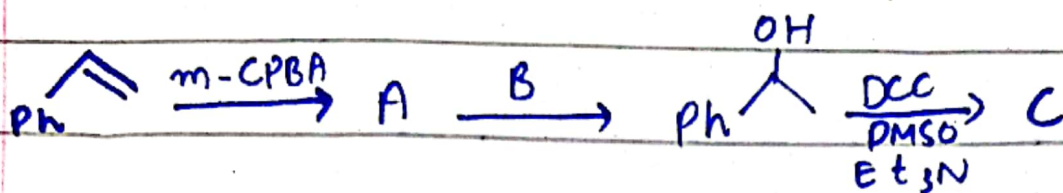
Regioselectivity

Hydroboration-oxidation is an anti-Markovnikov reaction. This means that the alcohol will end up on the less substituted carbon, while the hydrogen

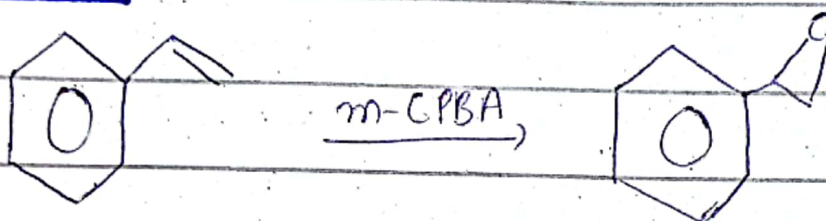
• adds to the more substituted carbon-



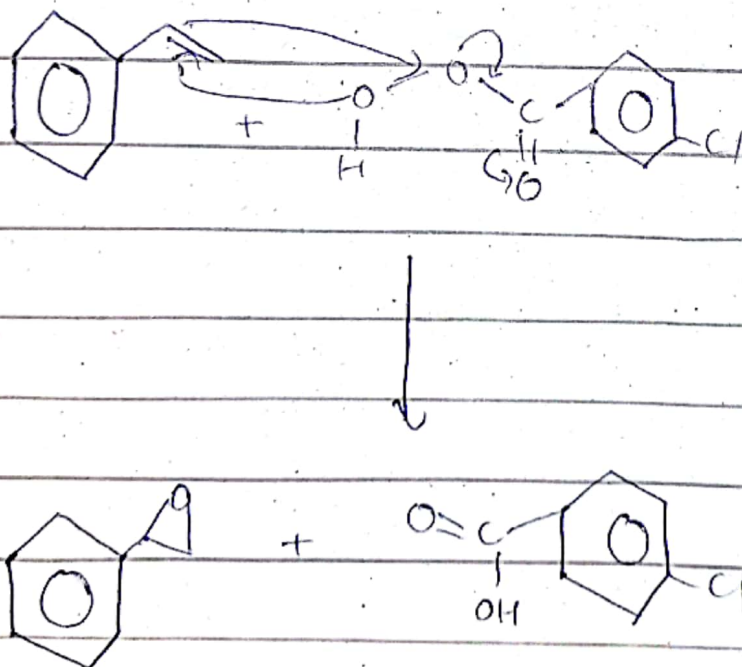
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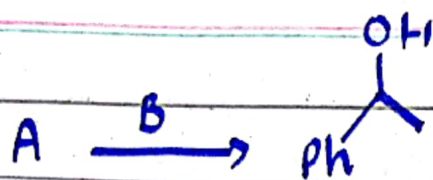


Reaction:

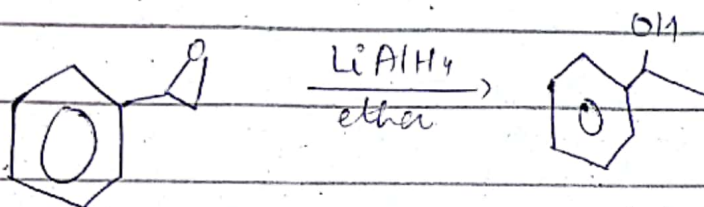


Mechanism



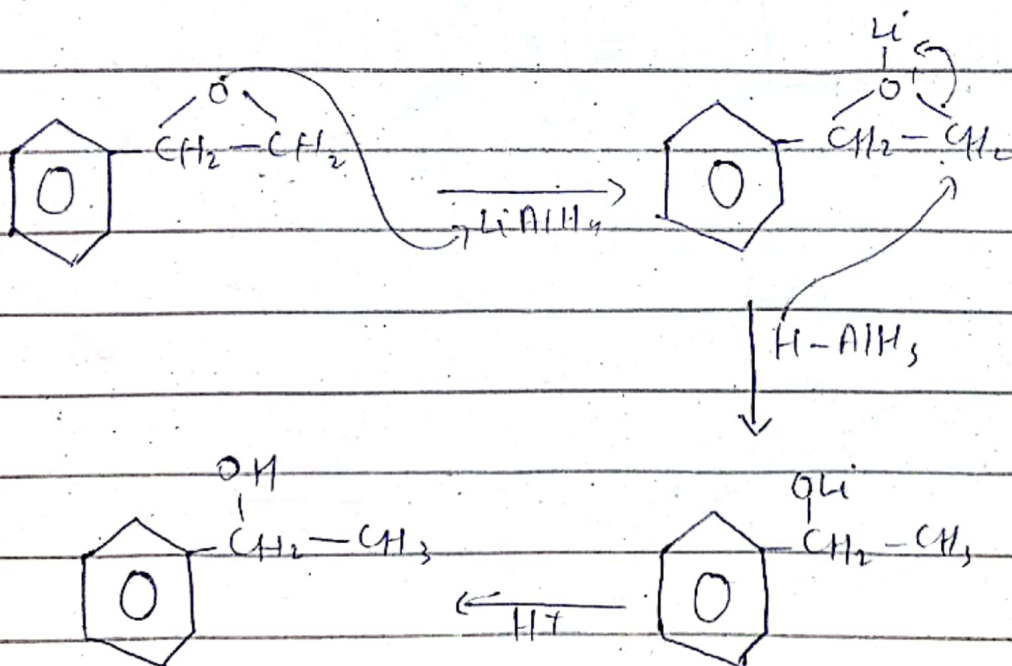
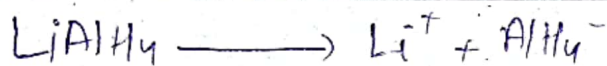


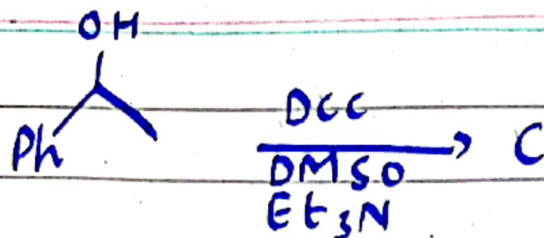
Reaction:



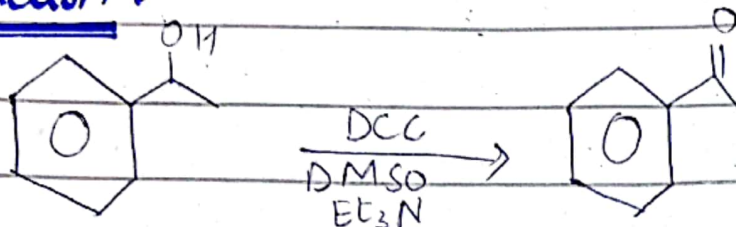
Epoxide convert into alcohol in the presence of LiAlH_4 and ether by reductive cleavage.

Mechanism





Reaction:



Secondary alcohol react with Dimethyl sulfoxide (DMSO) and N,N-dicyclohexylcarbodiimide (DCC) in the presence of triethylamine (Et_3N).

Mechanism:

